Math 351
Second Exam

Name:_____

Signature:_____

1. Answer all questions.
2. Show **ALL** your work in the space provided. If work is continued on the back of any test page, clearly indicate its presence.
3. All University rules and guidelines for student conduct are applicable.
4. Calculators with algebraic, calculus or matrix functions may not be used.
5. There are six problems on seven pages (this page and six others).

Results:

1. out of 16 points you got _____
2. out of 16 points you got _____
3. out of 16 points you got _____
4. out of 16 points you got _____
5. out of 20 points you got _____
6. out of 16 points you got _____

Your total: _____ out of 100

1. Let

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 3 & 3 & 4 \\ 2 & 2 & 3 \end{pmatrix}$$

Find A^{-1} and use it to solve

$$\begin{pmatrix} 1 & 0 & 1 \\ 3 & 3 & 4 \\ 2 & 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

2. Are

$$\operatorname{span}\left\{\begin{pmatrix}1\\3\\1\end{pmatrix},\begin{pmatrix}2\\-1\\1\end{pmatrix}\right\} \text{ and } \operatorname{span}\left\{\begin{pmatrix}3\\2\\2\end{pmatrix},\begin{pmatrix}4\\-3\\2\end{pmatrix}\right\}$$

the same or not? Why?

3. For what values of a does $Ax = 0$ have [illegible] solution?

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 2 & 0 & a \end{pmatrix}.$$

4. Let

$$T = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} \right\}.$$

Is the given set of vectors a basis for T and why? What is the dimension of T? If the set is not a basis, find a basis for T.

5. (Hint: these questions can be answered without calculation by appealing to theorems from the class)

(a) If A is a 7×5 matrix, why can the rank of A not be 6?

(b) Is

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 7 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} \right\}$$

a basis for $\mathbb{R}^3$ (why)?

(c) Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ 1 & 4 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & -1 & 4 \\ -1 & 3 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \text{ so } AB = \begin{pmatrix} -2 & 5 & 0 & 3 \\ 0 & 10 & 2 & 9 \\ -3 & 12 & 3 & 4 \\ 1 & 4 & 1 & 5 \end{pmatrix}$$

Find $B^T A^T$?

6. Expand the vector

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

in terms of the basis

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ -1 \\ -5 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right\}.$$